Great—this is exactly the right mindset 👍
Let’s go **deep but still easy to understand**, topic by topic, like your training module.

🐧 1. Introduction to Linux (Detailed)

Linux is an **Operating System (OS)** that:

- Manages hardware (CPU, memory, devices)
- Runs applications
- Allows user interaction (GUI + CLI)

🔹 Why Linux is Important

- Used in **servers (90%+)**
- Used in **cloud (AWS, Azure, etc.)**
- Base of **Android OS**
- Preferred by developers (fast + secure)

🔹 Key Features

- Multi-user → many users at same time
- Multitasking → many programs simultaneously
- Secure → strong permission system
- Open-source → free to modify

📜 2. Linux History & Philosophy

🔹 Origins of GNU/Linux

- **Linux Kernel** created by Linus Torvalds (1991)
- GNU Project provided tools (compiler, shell, etc.)
- Together → **GNU/Linux**

👉 Kernel = core brain of OS
👉 GNU tools = make it usable

🔹 Philosophy of Linux

- “Everything is a file”
- Small programs → do one thing well
- Combine commands for powerful tasks

🔹 Open Source Movement

- Source code is **public**
- Anyone can:
  - View
  - Modify
  - Distribute

💡 Example:
Companies like Google, Facebook use Linux but customize it

💻 3. Popular Linux Distributions

A **distribution (distro)** = Linux kernel + tools + UI

🔹 Ubuntu

- Beginner-friendly
- Large community
- Used in training & development

🔹 Fedora

- Latest features
- Used for testing new technologies

🔹 CentOS (or Rocky Linux now)

- Stable
- Used in companies/servers

🖥️ 4. Desktop Environments (GUI)

This is what you “see” on screen.

🔹 GNOME

- Default in Ubuntu
- Simple & clean

🔹 KDE

- Highly customizable
- More features

🔹 XFCE

- Lightweight
- Best for low RAM PCs

⌨️ 5. Command Line Basics

🔹 What is Terminal?

A program where you type commands:

```
user@system:~$
```

🔹 Shell

- Command interpreter
- Example: **Bash (most common)**

📂 6. Terminal Fundamentals

🔹 Access Terminal

- Shortcut: `Ctrl + Alt + T`

🔹 Basic Command Format

```
command [options] [arguments]
```

Example:

```
ls -l /home
```

📍 7. Basic Navigation Commands (Very Important)

🔹 **`pwd`**

Print current directory

```
pwd
```

🔹 **`ls`**

List files

```
ls
ls -l   (details)
ls -a   (hidden files)
```

🔹 **`cd`**

Change directory

```
cd folder
cd ..
cd ~   (home)
cd /   (root)
```

🗂️ 8. Directory Structure (Very Important)

Linux structure is like a tree:

```
/
├── home
├── bin
├── etc
├── root
├── var
```

🔹 **Important Directories**

- `/` → root (top level)
- `/home` → user files

- `/root` → admin user
- `/bin` → system commands
- `/etc` → config files
- `/var` → logs

📏 9. Absolute vs Relative Path

🔹 Absolute Path

Full path from root:

```
/home/soumi/file.txt
```

🔹 Relative Path

From current location:

```
file.txt
../file.txt
```

💡 Important for exams!

📁 10. File Management Commands (Core Topic)

🔹 Create Files

```
touch file.txt
```

🔹 Create Directory

```
mkdir folder
```

🔹 List Files

```
ls
```

🔹 Copy Files

```
cp file.txt backup.txt
```

🔹 Move / Rename

```
mv file.txt newname.txt
```

🔹 Delete File

```
rm file.txt
```

🔹 Delete Directory

```
rmdir folder
rm -r folder   (force delete)
```

⚠ Be careful with `rm -r` (permanent delete)

🔐 11. Bonus (Very Important for Interview)

🔹 File Permissions

```
-rwxr-xr-x
```

- r → read
- w → write
- x → execute

Change permission:

```
chmod 777 file.txt
```

🎯 Final Understanding (Wipro Exam View)

Focus MOST on:
✔ Commands (`ls`, `cd`, `rm`, `mkdir`)
✔ Directory structure
✔ Paths (absolute vs relative)
✔ Basic concepts (Linux, distro, shell)

💡 Smart Study Plan (For You)

Day 1:

- Learn commands + practice

Day 2:

- Directory + paths

Day 3:

- File management

Day 4:

- Mock test

This guide provides a detailed overview of the core concepts of **Linux Basics and the Command Line**, structured according to the provided syllabus.

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1. Introduction to Linux

Linux History and Philosophy

- **Origins of Linux & GNU/Linux:** Linux began in 1991 when Linus Torvalds, a Finnish student, created a free operating system kernel. It is often referred to as **GNU/Linux** because it combines the Linux kernel with the system tools and libraries developed by the GNU Project.
- **Open Source Movement:** Linux is the cornerstone of the open-source movement, which advocates for software that anyone can inspect, modify, and enhance. This philosophy promotes transparency, community collaboration, and rapid security updates.

Popular Linux Distributions (Distros)

Because Linux is open-source, different "flavors" or distributions exist to suit various needs:

- **Ubuntu:** Highly user-friendly and widely used for personal desktops and cloud servers.
- **Fedora:** Known for being cutting-edge, often introducing the latest features and software versions first.
- **CentOS:** Traditionally used for enterprise servers due to its stability and long-term support (often associated with Red Hat Enterprise Linux).

Understanding Desktop Environments

A Desktop Environment (DE) provides the Graphical User Interface (GUI) for Linux:

- **GNOME:** A modern, clean, and intuitive interface used by default in Ubuntu and Fedora.
- **KDE Plasma:** Highly customizable and resembles a traditional Windows-style layout.
- **XFCE:** Lightweight and fast, ideal for older hardware or users who prioritize performance over visual effects.

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2. Command Line Basics

Terminal Fundamentals

- **Accessing the Terminal:** The terminal (or shell) is the text-based interface used to interact directly with the operating system. It can be accessed via keyboard shortcuts (like `Ctrl+Alt+T` in many distros) or through the application menu.
- **Basic Navigation Commands:**
  - `pwd`: **Print Working Directory –** Shows exactly where you are in the system.
  - `cd`: **Change Directory –** Used to move between folders (e.g., `cd Documents`).

- `ls`: **List –** Shows the files and folders inside the current directory.

Directory Structures

Linux uses a hierarchical, tree-like file system starting from the **root**:

- **Key Directories:**
  - `/`: The **Root** directory; the starting point of the entire file system.
  - `/home`: Contains personal folders for users (e.g., `/home/username`).
  - `/bin`: Contains essential binary files (executable programs like `ls` or `cp`).
- **Absolute vs. Relative Paths:**
  - **Absolute Path:** The full path starting from root (e.g., `/home/user/Desktop`).
  - **Relative Path:** A path relative to your current location (e.g., if you are in `/home`, the relative path to Desktop is simply `user/Desktop`).

File Management Commands

These commands are used to manipulate files and folders:

- **Creating:**
  - `touch filename`: Creates a new empty file.
  - `mkdir foldername`: Creates a new directory (folder).
- **Listing:**
  - `ls -l`: Lists files with detailed information (permissions, size, date).
  - `ls -a`: Shows hidden files (those starting with a dot).
- **Deleting:**
  - `rm filename`: Removes a file.
  - `rm -r foldername`: Recursively removes a directory and all its contents.